▶ Present

DESIGNING LARGE-SCALE AI SYSTEMS · PROF. FABIO CASATI

CinePal

A conversational movie clustering system

TEAM CANTUCCI

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Live Demo

Video not bundled — watch it here:

github.com/user-attachments/assets/95a79f0f-2e41-40f0-88f5-f9e8a581a8bf

PREMISE

“How do you approach the conversational perspective?”

01

Operations

Free-form natural language is mapped onto a closed vocabulary of operations.

02

Snapshots

Every operation produces a snapshot. The oracle navigates and branches between them.

03

Satisfaction

Acceptance from the oracle is the objective function.

REPRESENTATION

Data Pipeline

Scrape TMDB, clean and split the catalogue, compute embeddings

01

Scrape

TMDB metadata, reviews and trailers

02

Clean & Split

Filter, deduplicate, normalise. dev / production / eval-holdout splits.

03

Embed

Three distinct modalities per film. Each with its own meaning.

REPRESENTATION

Movie Embedding

MODEL	BAAI / bge-large-en-v1.5
INPUT	Title, synopsis, genres, cast, director, keywords
OUTPUT	1024-d float32, L2-normalised
ROLE	The default semantic substrate

REPRESENTATION

Review Embedding

MODEL	BAAI / bge-large-en-v1.5
INPUT	Aggregated text of the top user reviews
OUTPUT	1024-d float32, L2-normalised
ROLE	Audience signal: plot details and tone

REPRESENTATION

Trailer Embedding

MODEL	OpenCLIP ViT-H/14
INPUT	Evenly-spaced trailer frames (fallback to posters)
OUTPUT	1024-d vector, mean-pooled across frames
ROLE	Visual modality: palette, pacing, style

REPRESENTATION

Fusion Strategies

Text and visual encoders inhabit different semantic spaces.

Vector-level

Movie and review embeddings lie in the same space.

Weighted mean of embeddings

Distance-level

Text and trailer need to be fused at the distance level.

Weighted sum of distance matrices

Passed to HDBSCAN as a **precomputed metric**

ARCHITECTURE

Coordinator

The deterministic orchestrator: receives the Intent Agent's output and decides what's next.

- The single source of truth for turn handling and snapshot handling
- Holds no in-memory session state between turns. Everything is persisted to the database.
- Takes the Intent Agent's ordered list of actions and routes workflow accordingly

ARCHITECTURE

Intent Agent

Interprets the oracle's free-form message and emits a structured, ordered list of actions.

- Each action carries a target cluster, parameters, and a confidence score
- Can resolve multiple actions in a single turn - e.g. "reset, then split by genre"
- Actions are executed in order; later steps see the result of earlier ones

INPUT

Message; Current cluster labels

OUTPUT

Ordered list of actions (params + confidence)

ARCHITECTURE

Operations Catalogue

A fixed, closed vocabulary, used by the Intent Agent.

CLUSTER
Split into sub-groups

MERGE
Combine clusters

FOCUS
Keep one, drop the rest

CROSS-FILTER
Match a predicate

EXCLUDE
Drop one cluster

RESET
Return to unclustered

UNDO
Step back a snapshot

EXPLAIN
Why a film sits here

SMALL-TALK
Non-operational reply

ARCHITECTURE

Concept Agent

Builds a linear axis between two poles; every film is projected onto it

- Triggered only for concept-guided requests
- Generates k sentences per pole, embeds them, and averages each set into a centroid
- Builds the linear axis from the negative to the positive centroid
- Each film's dot product with the axis gives its score

INPUT

Concept string; Declared space

OUTPUT

Linear axis (unit vector)

EXAMPLE · MOVIE ENDINGS

+ open, ambiguous endings
– closed, resolved endings

ARCHITECTURE

Clustering

The deterministic computational engine behind every clustering operation.

- A pure pipeline: UMAP reduction followed by HDBSCAN
- Returns per-film membership probabilities, not rigid hard labels

ARCHITECTURE

Labeling Agent

A fast-model agent that names and describes clusters

- IF **Metadata partitions** → assigned deterministically
- IF **Semantic clusters** → assigned through a LLM call
- Anchored on the previous turn's names
- Always generates descriptions through a LLM call

INPUT

Cluster Exemplar movies; Label context

OUTPUT

Label context; Label; Description

ARCHITECTURE

Explanation Agent

A strong-model agent that explains why a given film sits in a given cluster.

- Grounded in the cluster's label and exemplars
- Outputs a concise, natural-language placement rationale

INPUT

Cluster label; Cluster summary; Exemplars

OUTPUT

Concise explanation

ARCHITECTURE

Clarifier Agent

A fast-model gate that asks one disambiguating question when confidence is low.

- Fires when any action falls below the confidence threshold
- Asks a single disambiguating question

INPUT

Low-confidence action; User message; Current cluster labels

OUTPUT

One clarifying question

ARCHITECTURE

Suggestion Agent

A fast-model responder that proposes a next step once the actions complete.

- Computes deterministic signals: a similar-cluster pair, a dominant cluster, a high noise fraction
- May decline if it cannot identify a coherent suggestion

INPUT

Current cluster; Signals; Last operation

OUTPUT

Proposed next step (or none)

INTERFACES

What We Expose

Three ways to meet CinePal: as an application, as an API, or as a tool-using agent.

01

HTTP API

Auth, conversations, snapshots, movies, concepts, evaluation

02

MCP Tools

Conversations, snapshots, members, movies, concept axes

03

Web UI

Chat, history, movie scatterplot, cluster inspect, evolution graph, concept axes, evaluation lab

EVALUATION

Evaluation Strategy

Does an intent-driven navigation loop produce more coherent clusterings than other variants?

01

Operation Choice

Does the system pick the right operation?

02

Cluster Strategy

Does our approach provide a coherent clustering?

03

Satisfaction

How do we assess quality and the oracle's satisfaction?

EVALUATION

The Two Conditions

We compare CinePal against the simplest credible alternative

Conversational

Our full system

The complete multi-agent loop

Baseline

Monolithic LLM

A single strong-model call per turn

ORACLE · GROUND TRUTH

Ground Truth Building

No intrinsic ground truth exists, so we build one for every session.

Trajectory

An ordered plan

A target sequence of navigation operations, each step tagged with its action and concept.

Intent Description

A neutral paraphrase

Voice-agnostic natural language generated by an LLM from the trajectory.

ORACLE · BEHAVIOURAL MODEL

Oracle Persona

Two dials layered on the ground truth control how the persona behaves.

Verbosity

How much it says

Sets prompt-length behaviour: terse, medium, or verbose.

Patience

How long it stays

How many misunderstandings the oracle tolerates before giving up.

EVALUATION · METRICS

Session Metrics

CLUSTER COUNT	Number of clusters in the final snapshot
TURNS	Number of oracle turns in the session
OPERATIONS	Number of navigation operations executed
CLARIFIER RATE	Fraction of turns on which the clarifier fired
TOTAL COST	Cumulative LLM cost of the session
ORACLE RATING	The oracle's own self-rating, 1–5

EVALUATION · METRICS

Judge Metrics

OPERATION APPROPRIATENESS	Right operation and parameters for each request
INTENT ALIGNMENT	Does the final result reflect the stated goal?
LABEL ACCURACY	Do names and summaries match their operations?
CLUSTERING COHERENCE	Do films within a cluster genuinely belong together?
EXPLANATION QUALITY	Is a placement explanation faithful and specific?
SUGGESTION MEANINGFULNESS	Is a volunteered suggestion relevant and well-timed?
CONCEPT-AXIS QUALITY	Do the axis poles form a coherent, discriminative spectrum?

MAJOR FINDINGS

Operation Appropriateness

Where the conversational loop is strongest.

Strength

Right operation, every turn

Picks the correct operation and parameters, on the oracle's trajectory.

Score

3.94 ± 0.31

Our strongest, most reliable dimension.

MAJOR FINDINGS

Abstract Concept Cliff

Where shallow embeddings run out of road.

The Cliff

Concepts our embeddings miss

Very abstract ideas live outside the embedding space, so the system cannot split on them.

- a film movement (New Hollywood)

The Bridge

Retrieval-augmented depth

For a concept, retrieve each film's scripts and reviews; an LLM scores the film on it directly (RAG).

- Costly retrieval + LLM per film

MAJOR FINDINGS

Cluster Segmentation

Opposite failure modes on structure.

Our system

A balanced handful

- Usually 4.14 ± 1.67 clusters
- Can over-segment to 17–18 groups due to the oracle

Baseline

Chronic under-segmentation

- ≤ 2 clusters in 87% of sessions
- Distinct films lost during the conversation

FINDINGS

Other Findings

01 Operational Appropriateness

02 Cost of the System

03 Cluster Segmentation

04 Abstract Concept Cliff

05 Explanation Problem

06 Suggestion Problem

07 Labelling Problem

08 Oracle & Judge Divergence

RETROSPECTIVE

What Was Hard

01 Maintainability

02 Intent Agent

03 Domain-agnostic catalogue

04 Oracle design

05 Labeling consistency

06 Managing context at scale

RETROSPECTIVE

Who Did What

Davide Donà

- Database Schemas
- Ingestion Pipeline
- Clustering Engine
- Agents Orchestration
- Agent Prompting
- Config & Logging
- Automated Testing

Andrea Blushi

- Web UI
- FastAPI
- MCP Server
- Deployment
- Auth System
- Ground Truth Builder
- Evaluation Setup

APPENDIX · CONFIGURATION

Parameters Configuration

EMBEDDINGS & FUSION	VALUE
Text / review encoder	BGE-large-en-v1.5
Trailer encoder	OpenCLIP ViT-H/14
Embedding dimension	1024
Text–review fusion (text / review)	0.6 / 0.4
Text–trailer fusion (text / trailer)	0.6 / 0.4
CLUSTERING	**VALUE**
HDBSCAN min. cluster size (base / online)	50 / 10
HDBSCAN min. samples	10
UMAP components / neighbours	50 / 15

INTERACTION	VALUE
Intent confidence threshold	0.6
Turn budget (max. oracle turns)	15
Exemplar titles per cluster	5
COST & MODELS	**VALUE**
Cost ceiling — system / oracle / judge (USD)	10 / 2 / 2
Decoding seed (all models)	42
System — strong / fast	GPT-4o / 4o-mini
Oracle / Judge	Gemini 2.5 / Opus 4.7

APPENDIX · QUANTITATIVE RESULTS

Judge Dimensions & Oracle Rating

DIMENSION (1–5)	CONVERSATIONAL	BASELINE
Operation appropriateness	3.94 ± 0.31	2.73 ± 0.35
Intent alignment	3.09 ± 0.43	2.33 ± 0.47
Label accuracy	3.03 ± 0.44	3.53 ± 0.36
Clustering coherence	2.54 ± 0.29	2.87 ± 0.31
Explanation quality	1.79 ± 0.31	n/a
Suggestion meaningfulness	2.71 ± 0.21	n/a
Concept-axis quality	3.48 ± 0.35	n/a
Oracle self-rating	3.41 ± 0.49	2.82 ± 0.64

Mean ± 95% CI · 35 conversational and 30 baseline sessions over a held-out catalogue of 288 films.

APPENDIX · QUANTITATIVE RESULTS

Session Metrics & Termination

SESSION METRIC	CONV.	BASE.
Final cluster count	4.14 ± 1.67	1.97 ± 0.53
Turns per session	13.34 ± 1.08	12.40 ± 1.27
Operations per session	9.83 ± 1.16	11.50 ± 1.30
Operations per turn	0.74	0.93
Clarifier trigger rate	0.23 ± 0.04	0.00
Cost per session (USD)	0.169	0.153
Cost per turn (USD)	0.0126	0.0123

TERMINATION OUTCOME	CONV.	BASE.
Finished, intent fulfilled	17/35 · 49%	11/30 · 37%
Stopped, misbehaviour	12/35 · 34%	16/30 · 53%
Budget exhausted	6/35 · 17%	3/30 · 10%

APPENDIX · ETHICS

Ethical Considerations

01

Synthetic oracle limits

All evaluation relies on an LLM-simulated oracle, so findings reflect agent-to-agent dynamics, not authentic human behaviour.

02

Model & embedding bias

The language and embedding models are trained predominantly on Western, English-language data, inheriting its cultural biases.

03

LLM-as-a-judge risks

Using an LLM to evaluate another introduces structural risks around accuracy, blind spots, and hallucination.

EVALUATION

Notes

01

Three LLM families

One family runs the system, one the simulated oracle, one the judge.

02

Confidence intervals

Every reported metric carries its two-sided 95% confidence interval.

03

Oracle model class

The oracle does not need a strong reasoning model to play its part.

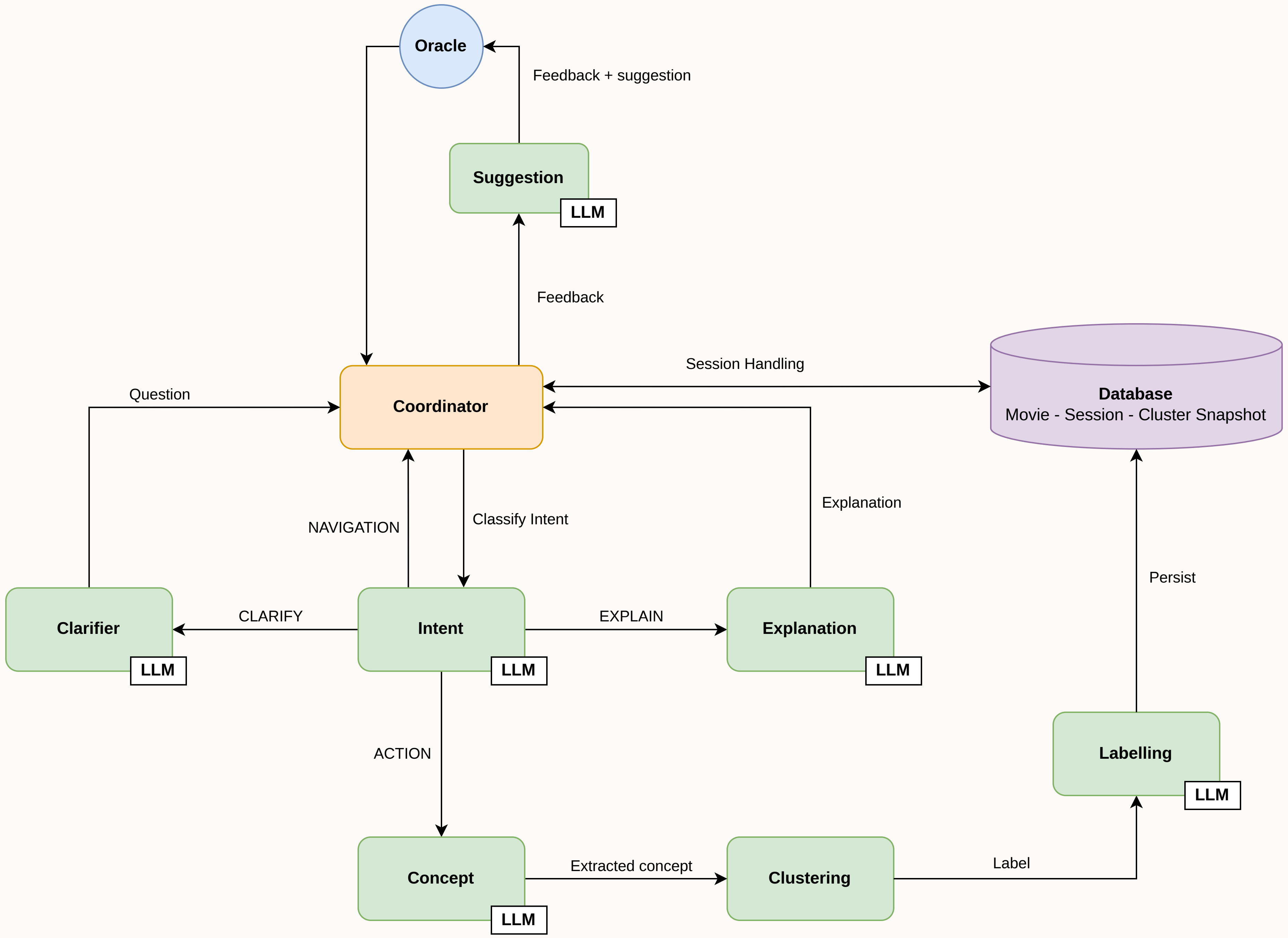
APPENDIX · EVALUATION

Confidence Interval

$$\mathrm{CI}_{95} = \bar{x} \pm t_{0.975,\, n-1} \cdot \frac{s}{\sqrt{n}}, \qquad s = \sqrt{\frac{1}{n-1} \sum_{i=1}^{n} (x_i - \bar{x})^2}$$

APPENDIX · DIAGRAM

System Architecture



APPENDIX · DIAGRAM

Turn Handling